

[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)
[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

H. Diamonds under W

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a 2D grid with R rows and H columns.

Each cell in the grid is one of the following:

1. $.$ — Empty cell: You can move into this cell.
2. D — Cell with a diamond: You can move into this cell and collect the diamond.
3. $\#$ — Obstacle: You cannot move into this cell.

You may start from any non-obstacle cell and move in the four directions: up, down, left, or right. Your goal is to choose a starting cell such that you can collect the maximum number of diamonds

Input

The first line contains two integers R and H ($1 \leq R, H \leq 1000$) — the number of rows and columns of the grid. The next R lines each contain H characters, describing the grid.

Output

Print a single integer — the maximum number of diamonds you can collect starting from a valid cell.

Examples

input

Copy

```
5 5
. # . D D
# # . # .
#####D
# D D D #
# . . D D
```

output

Copy

```
5
```

input

Copy

```
5 5
D #####
# # . D #
# . . D #
#####D
. . # # D
```

output

Copy

```
3
```

input

Copy

```
5 5
.....
####.
#..#.
####.
.....
```

CSE221 Assignment 05 Summer 2026

Contest is running

4 weeks

Contestant



→ Submit?

Language: Java 21 64bit

Choose file:

Choose File

No file chosen



Success!



Submit

output

Copy

0

input

Copy

1 5
D....**output**

Copy

1

input

Copy

9 11
.#..D...D..
.#.#####.
D#.#..D...#.
D#D#..###.#D
.#.#..D#.#.
.#.#####.#D
D#..D...D#..
.#####.
...D..D...D**output**

Copy

15

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Jul/19/2026 14:43:30^{UTC+6} (i1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by

